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AArch64 Memory & Cache System — Technical Reference Workbook

Chapter 7 — MMIO, Device-nGnRnE, Ordering Rules, and Barrier-Only Visibility (Based on ARM64 Cortex-A72)

7.1 Scope

This chapter defines the architectural behavior of **MMIO (Memory-Mapped I/O)** and **Device-nGnRnE** regions on ARM64 Cortex-A72. It explains:

- Why MMIO must not be cached
- Why DC/IC operations are architecturally ignored
- How ordering is enforced using barriers
- How Device memory interacts with PoC
- How CPU, DMA, and peripherals observe MMIO
- Correct usage patterns for register access
- Required barrier semantics for safe device communication

This chapter is essential for firmware, bootloader, and driver development.

7.2 Device Memory Types

ARMv8-A defines several Device memory types. The most important for MMIO is:

7.2.1 Device-nGnRnE (Non-Gathering, Non-Reordering, Non-Early-Write-Acknowledgement)

Characteristics:

- **Non-cacheable**
- **Strongly ordered**
- **No speculation**
- **No merging or combining**
- **No write buffering**
- **No DC/IC effects**
- **No prefetching**
- **No reordering across accesses**

This is the correct attribute for:

- UART registers
- SPI/I2C controllers
- Interrupt controllers
- Timers
- SoC control registers
- Peripheral MMIO windows

Incorrectly marking MMIO as Normal memory results in undefined behavior.

7.3 Why MMIO Must Not Be Cached

Caching MMIO is architecturally illegal because:

1. **Registers change independently of CPU caching**
2. **Reads must always reflect the latest device state**
3. **Writes must be observed by the device immediately**
4. **Speculation must not occur**
5. **Prefetch must not occur**
6. **Write combining must not occur**
7. **Reordering must not occur**

Caching MMIO breaks device protocols and can corrupt hardware state.

7.4 DC/IC Operations on Device Memory

7.4.1 Architectural Rule

All DC and IC operations are **ignored** for Device-nGnRnE memory.

7.4.2 Implications

- DC CVAC/CIVAC/CVAU have no effect
- IC IVAU/IALLU have no effect
- No dirty lines exist
- No invalidation is required
- No writeback is required

Device memory is always visible at PoC.

7.5 Ordering Rules for MMIO

Device memory requires strict ordering. ARM defines:

7.5.1 No Reordering Across Device Accesses

Loads and stores to Device-nGnRnE cannot be reordered relative to each other.

7.5.2 Reordering Relative to Normal Memory

Device accesses may be reordered relative to Normal memory unless barriers are used.

7.5.3 Required Barriers

Use:

- **DSB** — enforce completion of prior memory operations
- **ISB** — enforce instruction pipeline synchronization
- **DMB** — enforce ordering without completion

Correct barrier usage is mandatory for device drivers.

7.6 Required Barrier Semantics

7.6.1 DSB (Data Synchronization Barrier)

Ensures:

- All prior memory accesses complete
- All device writes reach the peripheral
- All device reads observe the latest state

Required after:

- Writing device control registers
- Writing DMA descriptors
- Writing interrupt controller registers

7.6.2 DMB (Data Memory Barrier)

Ensures ordering but not completion.

Used when:

- Ordering is required
- Completion is not required

7.6.3 ISB (Instruction Synchronization Barrier)

Required when:

- Device registers affect instruction execution
- Changing system control registers
- Modifying MMU configuration

7.7 MMIO Access Patterns

7.7.1 Write to Device Register

Required sequence:

```
STR  wX, [device_reg]
DSB  SY
```

Ensures the device observes the write.

7.7.2 Read from Device Register

Required sequence:

```
DSB  SY
LDR  wX, [device_reg]
```

Ensures the read observes the latest device state.

7.7.3 Writing DMA Descriptors

Required sequence:

```
STR  wX, [descriptor]
DSB  SY
STR  wY, [device_start_register]
DSB  SY
```

Ensures:

- Descriptor is visible
- Device sees the start command

7.7.4 Reading DMA Completion Status

Required sequence:

```
DSB  SY
LDR  wX, [device_status]
```

Ensures CPU observes the latest status.

7.8 Device Memory and PoC

Device memory is always visible at PoC:

- No caching
- No dirty lines
- No writeback
- No invalidation
- No DC/IC operations

PoC visibility is immediate.

7.9 Microarchitectural Notes (Cortex-A72)

Although hidden behind architectural rules:

- Device memory bypasses L1D and L2
- Accesses go directly to interconnect
- Ordering is enforced by barrier units
- No speculation or prefetch occurs
- No FEQ involvement
- No snoop involvement

These behaviors ensure deterministic device access.

7.10 Summary Table

Feature	Normal Cacheable	Device-nGnRnE
Caching	Yes	No
Dirty lines	Yes	No
DC operations	Meaningful	Ignored
IC operations	Meaningful	Ignored
Speculation	Allowed	Forbidden
Prefetch	Allowed	Forbidden
Reordering	Allowed	Forbidden
Visibility	PoU/PoC	Always PoC
Barriers	Required for correctness	Required for ordering

7.11 Exercises (Technical Reference)

Exercise 7.1 — Device Memory Attributes

Explain why Device-nGnRnE must be used for MMIO regions.

Exercise 7.2 — DC/IC Semantics

Describe why DC CVAC and IC IVAU have no effect on Device-nGnRnE memory.

Exercise 7.3 — Ordering

Explain why DSB is required after writing device control registers.

Exercise 7.4 — DMA Descriptor Visibility

Describe the required sequence to ensure DMA sees updated descriptors.

Exercise 7.5 — Speculation

Explain why speculative loads must not occur on MMIO regions.